

Xiangyan Liu

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Education

National University of Singapore <i>Ph.D.</i> in Computer Science, School of Computing Advised by Michael Qizhe Shieh	Aug. 2024 - Present Singapore
National University of Singapore <i>M.Sc.</i> in Statistics, Department of Statistics and Data Science	Aug. 2022 - Jan. 2024 Singapore
Shanghai University of Finance and Economics <i>B.Sc.</i> in Data Science, School of Statistics and Management	Sep. 2018 - Jun. 2022 Shanghai, China

Research Interests

My current research centers on multimodal and agentic foundation models, with an interest in solutions that are simple, generalizable, and scalable.

Experience

† National University of Singapore, Tree and Rock AI Lab <i>Ph.D. student</i> , advised by Michael Qizhe Shieh	Singapore Aug. 2024 – Present
- Co-led a research project stress-testing language models' MCP use in complex contexts, with strict state tracking (initialization, isolation, reset) and automated verification scripts to ensure safe, reliable evaluation. - Contributed to a research project proposing a scalable data synthesis pipeline that improves VLM reasoning by generating challenging, verifiable data for RLVR training, yielding strong gains on hard benchmarks.	
Sea AI Lab <i>Associate member</i> , working with Tianyu Pang , Chao Du , and Longxu Dou	Singapore Oct. 2024 – Present
- Led a research project on improving VLM reasoning by developing a data augmentation technique that mixes clean and noisy trajectories during RL training to enhance reasoning robustness—with no extra cost. - Contributed to a research project on Next-Event Prediction, a self-supervised task that improves video LLMs' temporal reasoning by predicting future events from past video segments. - Contributed to a research project on feedback-conditional policy (FCP), which reframes feedback-driven learning as conditional generation, enabling LLMs to directly learn from verbal feedback.	
Moonshot AI <i>Research Intern</i> , working with Yiping Bao and Jinguo Zhu	Singapore Sep. 2025 – Present
- Contributed to the OK Computer project on improving website generation with iterative self visual critics - Contributed to the post-training data construction of Kimi-K2.5	
Alibaba Group, Tongyi Lab <i>Research intern</i> , working with Wenmeng Zhou	Hangzhou, China Mar. 2024 – Aug. 2024
Led a research project on building a code agent system that uses graph databases as interfaces to enhance LLM interaction with code repositories, enabling precise code retrieval and navigation	
WestLake University, LINs Lab <i>Visiting student</i> , advised by Tao Lin	Hangzhou, China May. 2023 – Nov. 2023
Led a research project on multimodal complex reasoning, proposing an efficient model selection method that captures dependencies among the sample, model, and subtask.	
National University of Singapore, NExT Research Centre <i>Research intern</i> , working with Wei Ji	Singapore (remote) Jan. 2023 – Nov. 2023
Co-led a research project on multimodal sequential recommendation, proposing ID-aware multimodal transformer and online distillation strategies to improve item representation learning and information fusion.	

Selected Publications (* denotes equal contribution, † denotes student lead)

MCPMark: A Benchmark for Stress-Testing Realistic and Comprehensive MCP Use

Zijian Wu^{†*}, Xiangyan Liu^{†*}, Xinyuan Zhang*, Lingjun Chen*, Fanqing Meng*, Lingxiao Du*, Yiran Zhao*, Fanshi Zhang*, Yaoqi Ye, Jiawei Wang, Yufan Yang, Jinjie Ni, Zirui Wang, Arvin Xu, Michael Qizhe Shieh
International Conference on Learning Representations (ICLR), 2026

NoisyRollout: Reinforcing Visual Reasoning with Data Augmentation

Xiangyan Liu*, Jinjie Ni*, Zijian Wu*, Chao Du, Longxu Dou, Haonan Wang, Tianyu Pang, Michael Qizhe Shieh
Advances in Neural Information Processing Systems (NeurIPS), 2025

CodexGraph: Bridging Large Language Models and Code Repositories via Code Graph Databases

Xiangyan Liu*, Bo Lan*, Zhiyuan Hu, Yang Liu, Zhicheng Zhang, Fei Wang, Michael Shieh, Wenmeng Zhou
North American Chapter of the Association for Computational Linguistics (NAACL), 2025

Towards Robust Multi-Modal Reasoning via Model Selection

Xiangyan Liu*, Rongxue Li*, Wei Ji, Tao Lin
International Conference on Learning Representations (ICLR), 2024

Online Distillation-Enhanced Multi-Modal Transformer for Sequential Recommendation

Wei Ji*, Xiangyan Liu*, An Zhang, Yinwei Weo, Yongxin Ni, Xiang Wang
ACM International Conference on Multimedia (ACM MM), Oral, 2023

Service

Conference Reviewer: ICML (2026), ICLR (2025, 2026), NeurIPS (2024, 2025), ACM MM (2023, 2024)

Skills

Programming Languages: Python, R, SQL, Shell Script, Scala, LaTeX

Frameworks and Tools: PyTorch (Distributed Training), Hugging Face, vLLM, Ray, Spark, Docker

Languages: Mandarin (native), English (fluent)

Teaching

Teaching Assistant

CS5260 - Neural Networks and Deep Learning II, National University of Singapore

Singapore
Spring 2025